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Concepts of IoT

Corrected lesson path for **Course Code 6042A**. This handbook explains IoT fundamentals, M2M, architecture, sensors, actuators, Arduino, NodeMCU, Raspberry Pi, communication protocols, cloud connectivity and domain applications.

Course Code
6042A

Correct file
lessons-6042A.html

Course overview

IoT means connecting physical things to digital systems using sensors, controllers, communication networks, cloud platforms and applications. The core engineering question is simple: what data must be sensed, how it is processed, where it is transmitted, and what action must happen.

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Area	What to learn
Sensing	Temperature, light, distance, motion, current, voltage and status signals.
Processing	Microcontroller or single-board computer reads data and makes decisions.
Communication	Wi-Fi, Bluetooth, serial links, MQTT, HTTP and cloud APIs.
Application	Monitoring, automation, alerts, dashboards, control and analytics.

Module 1 - IoT fundamentals

Definition

Internet of Things is a network of physical objects embedded with sensors, electronics, software and connectivity so that they can collect, exchange and act on data.

IoT vs M2M

M2M is machine-to-machine communication, often point-to-point. IoT is broader and usually includes internet/cloud integration, analytics, dashboards and scalable device management.

Basic architecture

Device layer → connectivity layer → data processing/cloud layer → application layer. Security, power management and maintenance are required across all layers.

Module 2 - IoT hardware platforms

Arduino

Good for simple sensor reading, digital/analog I/O and actuator control. It is easy for beginner embedded experiments.

NodeMCU / ESP8266

Microcontroller board with Wi-Fi. Suitable for low-cost network-connected sensor nodes and control projects.

Raspberry Pi

Single-board computer running Linux. Suitable when camera, database, web server or heavier processing is needed.

Sensors and actuators

Sensors convert physical quantities into electrical data. Actuators convert control signals into action, such as relay switching, motor movement or indication.

Module 3 - Communication and protocols

Wi-Fi

Common for internet access where power and network coverage are available.

Bluetooth

Used for short-range personal/device communication.

MQTT

Lightweight publish/subscribe protocol. It uses broker, publisher, subscriber and topic.

HTTP

Request/response protocol used for web APIs and dashboards.

Data flow example

Temperature sensor reads machine room temperature → NodeMCU sends value by MQTT → cloud dashboard displays trend → alert is generated if the value crosses limit.

Module 4 - Applications and engineering use

Smart home

Lighting, security, energy monitoring and appliance control.

Industry

Condition monitoring, predictive maintenance, production tracking and alarms.

Healthcare

Wearables, remote monitoring and equipment status logging.

Escalator/electrical relevance

IoT can monitor controller status, temperature, energy use, run hours and fault alarms, but safety-critical control must follow approved industrial standards.

Question bank

1. Define IoT and explain its architecture.
2. Compare IoT and M2M communication.
3. Explain sensors and actuators used in IoT.
4. Compare Arduino, NodeMCU and Raspberry Pi.
5. Explain MQTT with publisher, broker, subscriber and topic.
6. Explain IoT applications in industry.
7. Describe security challenges in IoT.
8. Design a basic IoT temperature monitoring system.

Model question paper

PART A: Define IoT, M2M, sensor, actuator, MQTT, cloud dashboard. PART B: Explain IoT architecture, NodeMCU features, Raspberry Pi applications, MQTT working, IoT security. PART C: Design an IoT monitoring system for an industrial machine and explain hardware, communication, dashboard and safety limitations.